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Can AI streamers drive purchases in food delivery E-commerce live streaming? Evidence from an O2O platform

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ABSTRACT

AI-enabled e-commerce live broadcast marketing has become a reality. This study explores the role of AI streamers in driving consumer purchases within the food delivery live-streaming sector. Using real-time behavioral data from the Meituan platform—a leading O2O food delivery service—we find that AI streamers significantly enhance live-stream performance by increasing the rate of purchase compared to human streamers. To uncover the mechanistic pathways through which AI streamers influence consumer behavior, we examine the moderating roles of linguistic emotionality and vocal dynamism. Results indicate that the positive impact of AI streamers on purchases is amplified when their language contains higher levels of emotional expressiveness and when their vocal delivery demonstrates greater variability in loudness. Additionally, while complex language styles tend to inhibit purchase behaviors, speech speed and pitch variability are crucial for sustaining consumer engagement. Visual simplicity in the live-stream background also plays a key role in maintaining audience attention. These findings provide actionable guidance for businesses on integrating AI streamers into food delivery live-streaming practices. Specifically, firms can refine the linguistic and vocal strategies of AI streamers to simulate human-like engagement and optimize consumer purchase behaviors. By leveraging these mechanisms, businesses can enhance the effectiveness of AI streamers in food delivery e-commerce, ultimately increasing both consumer engagement and sales performance.

1. Introduction

The rapid evolution of digital marketing has seen live-streaming e-commerce emerge as a novel and influential marketing channel (Guo et al., 2025). This channel not only offers a real-time interactive shopping experience but also caters to consumers' social and emotional needs. As a result, live-streaming e-commerce has become a powerful tool for driving sales, building brand awareness, and fostering customer loyalty. The application of Artificial Intelligence technology presents new opportunities for live-streaming e-commerce (Chen and Zhang, 2025). These virtual streamers can work around the clock to deliver consistent and engaging content without taking breaks.[1] In addition, AI streamers can be tailored to appeal to specific groups and products, enabling companies to target their marketing more effectively (Puntoni et al., 2021). While this channel successfully addresses consumers' need for sensory immersion (e.g., real-time product demonstrations), it also faces an operational dilemma: human streamers' emotional labor cannot sustainably scale, whereas AI streamers struggle to replicate authentic persuasion (Huang and Rust, 2021; Zhao et al., 2025; Zhou et al., 2024). Improving consumer engagement and purchases has become more complex, requiring the strategic deployment of AI streamers to enhance marketing effectiveness.

However, despite the growing importance of AI in this domain, research on its effects in live streaming, particularly its impact on sales performance, remains limited (Zhou et al., 2024). Current research mainly focuses on two aspects: one part of the research reveals the differences between AI and real people in influencing consumer emotions and trust (Gong and Sun, 2025; Yin et al., 2024; Zhang and Wang, 2024); the other part of the research emphasizes the efficiency of artificial intelligence in reducing costs and providing 24/7 availability (Huang and

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[1] Meituan Live's GMV exceeded 2 billion, and some businesses relied on digital humans to achieve 24-h continuous broadcasting! https://news.sohu.com/a/741924179_121060424.

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Rust, 2022; Wang et al., 2025; Yuan et al., 2025). Thus, a critical gap exists in understanding the strategic balance between technological capability and psychological acceptability, a balance that is pivotal for AI's sustainable integration into live-streaming commerce.

This study addresses this gap by examining whether AI streamers can, in fact, enhance the effectiveness of e-commerce live broadcasts. We comprehensively assess 13 variables influencing the purchase volume in e-commerce retail live broadcasts, considering factors such as streamer language and voice characteristics, streamer attributes, and the background features of the live broadcast environment. Drawing on a dataset from the Meituan platform, where live broadcasts with both human and AI streamers operated under the same brand, our empirical analysis reveals that live broadcast rooms employing AI streamers exhibit significantly higher purchase volumes than those with human streamers within the catering industry. This finding stands in stark contrast to most existing literature, which relies on situational psychological experiments and self-reported questionnaires (see Table 1) rather than actual observational data.

Furthermore, our subsequent analysis uncovers the key mechanisms through which AI streamers can effectively drive consumer purchases. This study leverages advanced tools, including deep learning and large language models, to systematically extract and analyze the linguistic, vocal, visual, and design features of live streams. We find that linguistic emotionality and vocal dynamism are crucial moderating variables. Specifically, the positive impact of AI streamers on purchase volume is significantly amplified when their language contains higher levels of emotional expressiveness and when their vocal delivery demonstrates greater variability in loudness. Our research further extends this understanding by exploring the nuanced effects of AI streamer appearance and gender design, as well as the visual simplicity of the live-stream background, revealing how these factors contribute to consumer engagement and purchase decisions.

By focusing on these comprehensive variables, this study aims to uncover the mechanistic pathways through which AI streamers simulate human-like engagement and influence consumer participation and purchasing during live streams. Ultimately, these findings offer empirical support for the strategic deployment of AI streamers to optimize retail marketing performance. First, regarding the design of AI streamer communication, brands can enhance effectiveness by incorporating emotionally expressive language and human-like vocal variations into AI scripting, thereby leveraging these mechanisms to increase consumer engagement and drive purchases. Second, our research provides

Table 1
Selected Literature on AI streamers in Retail.

References	Methods	Sample	Main DV(s)	Main IV(s)	CV(s)	Key findings
Li et al. (2023)	Scenario-based survey	289 Participants	Empathy level, call duration	AI vs. Humans	Gender, age, education level	Consumers show significantly less empathy for AI telemarketers than humans, leading to faster hanging up.
Zhang et al. (2023)	Scenario-based survey	559 Participants	Interaction Intention, Buy Impulsively	Perceived Coolness, Perceived Congruence, Perceived Mind Perception	Arousal, Impulsiveness, Mindset	Coolness directly affected parasocial interaction intention, while congruence and mind perception directly influenced the urge to buy impulsively.
Zhang and Wang (2024)	Scenario-based survey	753 Participants	Purchase intention	Virtual anchor type (Animal humanoid vs. full humanoid); Product type (practical vs. hedonic); Demand certainty (high vs. low)	Age, gender, income, product familiarity, brand information, and live broadcast scene consistency	Perceived warmth boosts purchase intention for hedonic products and uncertain needs—making animal-like virtual anchors more effective.
Xie et al. (2024)	Secondary Data Analysis, Scenario-based survey	10,000 live comments and 1469 Participants	Consumer Brand Forgiveness	Real human streamers vs. human-powered virtual streamers; brand reputation; streamer influence	Brand familiarity, host familiarity, and perceived situational authenticity.	Brands with high reputation or high influence are more suitable to adopt human-powered virtual streamers to reduce the negative impact of inappropriate speech.
Liu et al. (2025)	Scenario-based survey	663 Participants	Purchase Intention, Consumer Trust, Consumer Engagement	Emotion type: Positive/ Negative; Shopping values: Hedonic/ Utilitarian	Age, gender, education level, monthly income	AI virtual anchors are only highly accepted among extreme hedonistic consumers, while utilitarian-oriented consumers' trust in them has significantly decreased.
Gong and Sun (2025)	Scenario-based survey	915 Participants	Consumer Intention to Follow Advice (CIFA) and mind perception.	Virtual streamers' language style (emotional vs. rational)	Gender, watching frequency of livestreaming	Emotional language enhances perceived agency and perceived experience, which in turn increases CIFA.
Wang et al. (2025)	Scenario-based survey	463 Participants	Purchase intention	Language arousal (high vs. low), Influencer type (virtual vs. human	Perceived quality, perceived novelty, perceived coolness, source likeability, mood	Human influencers using low-arousal language evoke stronger purchase intentions. Virtual influencers using high-arousal language evoke stronger purchase intentions.
This study	Secondary data Analysis, Deep learning, text/ speech/image analysis	6 brands on Meituan, each with a real and AI live room (12 total), 480 min per room, 5760 samples.	Purchases per minute	AI streamer or human streamer、Linguistic emotionality and Vocal dynamism	Linguistic characteristics of streamers; Voice characteristics of streamers; Physical characteristics of streamers; Background characteristics of the streaming rooms; Number of followers of live room.	AI streamers boost catering sales more than humans, especially when their language exhibits high linguistic emotionality and their vocal delivery demonstrates significant vocal dynamism.

actionable guidance on the appearance and gender design of AI streamers and the visual strategy for the live-streaming background (Pan et al., 2021). We demonstrate that subtle yet impactful visual elements, such as background simplicity and color contrast, are crucial for minimizing distractions and directing consumer attention to the product. Our findings also offer insights into how the gender and aesthetic design of an AI streamer should be strategically aligned with the target audience to optimize viewer connection and purchase outcomes. By offering a comprehensive examination of AI streamers' linguistic, vocal, visual, and design-related attributes, this study provides a deep-dive into the application of AI streamers in food delivery e-commerce live streaming, bridging the gap between theoretical understanding and practical implementation.

2. Literature review

Live e-commerce has fundamentally reshaped how consumers interact with brands. In this context, the contrasting characteristics of AI and human streamers have sparked scholarly debates on whether—and under what conditions—AI can effectively promote marketing outcomes. Early research underscores the unique trust advantages of human streamers, stemming from their ability to deliver spontaneous interactions (e.g., unplanned promotional cues) and rich multimodal emotional expressions (e.g., voice modulation, micro-expression control), which help compensate for the sensory limitations of online channels. In food and beverage livestreams in particular, human streamers often leverage sensory-simulation language (e.g., "the aroma of crispy fried chicken is wafting") to trigger consumers' cross-sensory experiences and shorten the impulse purchase decision cycle (Zhou et al., 2024). This mechanism is especially critical in trust-intensive categories (e.g., luxury goods) (Gong et al., 2026; Jung et al., 2025; Lv et al., 2025), where emotional endorsement from the streamer helps mitigate perceived risks (Cascio Rizzo et al., 2023).

However, the strengths of human streamers are offset by inherent performance variability, such as fluctuations in emotional delivery and inconsistent presentation quality, which limits scalability. This challenge opens the door for AI streamers as a scalable alternative, especially in the food and beverage sector where purchases are often impulsive and decisions are made rapidly. AI streamers excel in standardized information delivery (e.g., ingredient details, time-limited promotions) thanks to their script control, fatigue-free operation, and 24/7 availability, reducing human errors like slips of the tongue or fatigue-induced monotony. However, this efficiency is context-dependent: in high-involvement purchases (e.g., large appliances), AI's mechanical language reveals semantic limitations, and complex structures (e.g., conditional clauses, metaphorical expressions) may overload cognitive processing, reducing purchase intentions. This tension can be understood through the Uncanny Valley effect—as AI approaches near-human simulation, subtle "inhuman" cues (e.g., repetitive speech rhythm) may trigger psychological discomfort and consumer rejection.

In this study, the term AI streamer refers to an algorithm-driven virtual host designed primarily for marketing and real-time sales communication within live-stream commerce platforms. Unlike virtual influencers, who typically possess independent digital identities, fictional backstories, and social media personas for long-term branding or entertainment purposes, AI streamers are functional agents embedded within e-commerce systems to deliver product information, interact with audiences, and stimulate purchases. Despite growing interest in AI streamers, several research gaps remain. Although prior studies have examined how linguistic, visual, or vocal cues influence consumer engagement in digital marketing and live-stream commerce (Wang et al., 2024; Zhang et al., 2023), most treat these modalities separately and rely on experimental or survey-based designs that capture attitudinal rather than behavioral outcomes. Real-time, high-frequency behavioral data—such as minute-by-minute purchasing patterns—are still scarce, limiting understanding of dynamic differences between AI and human streamers. Moreover, the synergistic impact of multimodal communication remains underexplored: while Seymour et al. (2024) investigated interface anthropomorphism, few studies have examined how language style complexity, pitch variation, and visual contrast interact to shape purchasing decisions. The mechanisms underlying AI streamer effectiveness also remain ambiguous. For example, Kirk and Givi (2025) found that emotionally charged AI content might reduce loyalty, yet the interplay between linguistic emotionality and vocal dynamism in driving real-time consumer behavior has not been tested. Addressing these gaps, the present study integrates linguistic, vocal, and visual data within a quasi-experimental design to provide a unified, behavior-based perspective on how multimodal communication cues drive consumer purchase behavior in AI-enabled live-streaming environments. To contextualize these gaps, Table 1 synthesizes representative studies on AI streamers in retail, outlining their methods, key variables, and findings, and highlighting how the present study addresses the identified limitations.

3. Theoretical background and hypothesis development

3.1. Virtual streamers and live streaming performance

The performance differences between AI and human streamers can be systematically explained through the integrated lens of CASA theory, Uncanny Valley Theory, and Information Processing Theory (IPT). CASA suggests that viewers often apply social heuristics to AI agents, allowing AI streamers to excel in standardized, repetitive, and accuracy-driven tasks—such as parameter-stable vocal delivery and fatigue-free product narration(Chang et al., 2025). This mechanical reliability helps AI streamers establish instrumental trust, especially in low-involvement, quick-decision contexts common in food delivery live commerce (Yuan et al., 2025).

However, Uncanny Valley Theory highlights the limits of AI persuasion: although AI's linguistic and vocal cues can be algorithmically optimized, AI streamers still struggle to reproduce the emotional authenticity, micro-expressions, and improvisational nuance that human streamers provide. Such emotional deficits become more salient in scenarios with higher perceived risk or symbolic value, where empathy, humor, and spontaneous interaction strengthen human streamers' social presence and trustworthiness (Li et al., 2023).

IPT complements these mechanisms by clarifying when AI or human cues dominate. AI streamers reduce cognitive load through consistent, clear information delivery, but they are less effective when consumers require rich contextual interpretation or emotional reassurance (Zhang et al., 2023). In contrast, human streamers perform better under analytic or trust-intensive processing conditions, where nuanced cues and adaptive communication matter more.

Overall, the integration of CASA, Uncanny Valley, and IPT reveals that AI and human streamers are not substitutes but complementary agents (Chen et al., 2025; Kirk and Givi, 2025; T.-G. Li et al., 2024; Sands et al., 2024). AI's strength lies in mechanical consistency and cognitive efficiency, while human streamers' advantage stems from emotional resonance and authenticity (Zhou et al., 2024). This complementary framework advances theoretical understanding by extending these classic theories to highly dynamic, commercial, and emotionally varied live-commerce environments.

By clarifying the boundary conditions and cross-context relevance of these mechanisms, the study positions AI anthropomorphism as a context-sensitive but theoretically generalizable construct. As shown in Figs. 1 and 2, based on the above theoretical and empirical perspectives, Hypothesis 1 is proposed.

H1. *AI streamers have better e-commerce live streaming performance than human streamers.*

To investigate the mechanistic pathways through which AI streamers enhance consumer purchases compared to human streamers, we

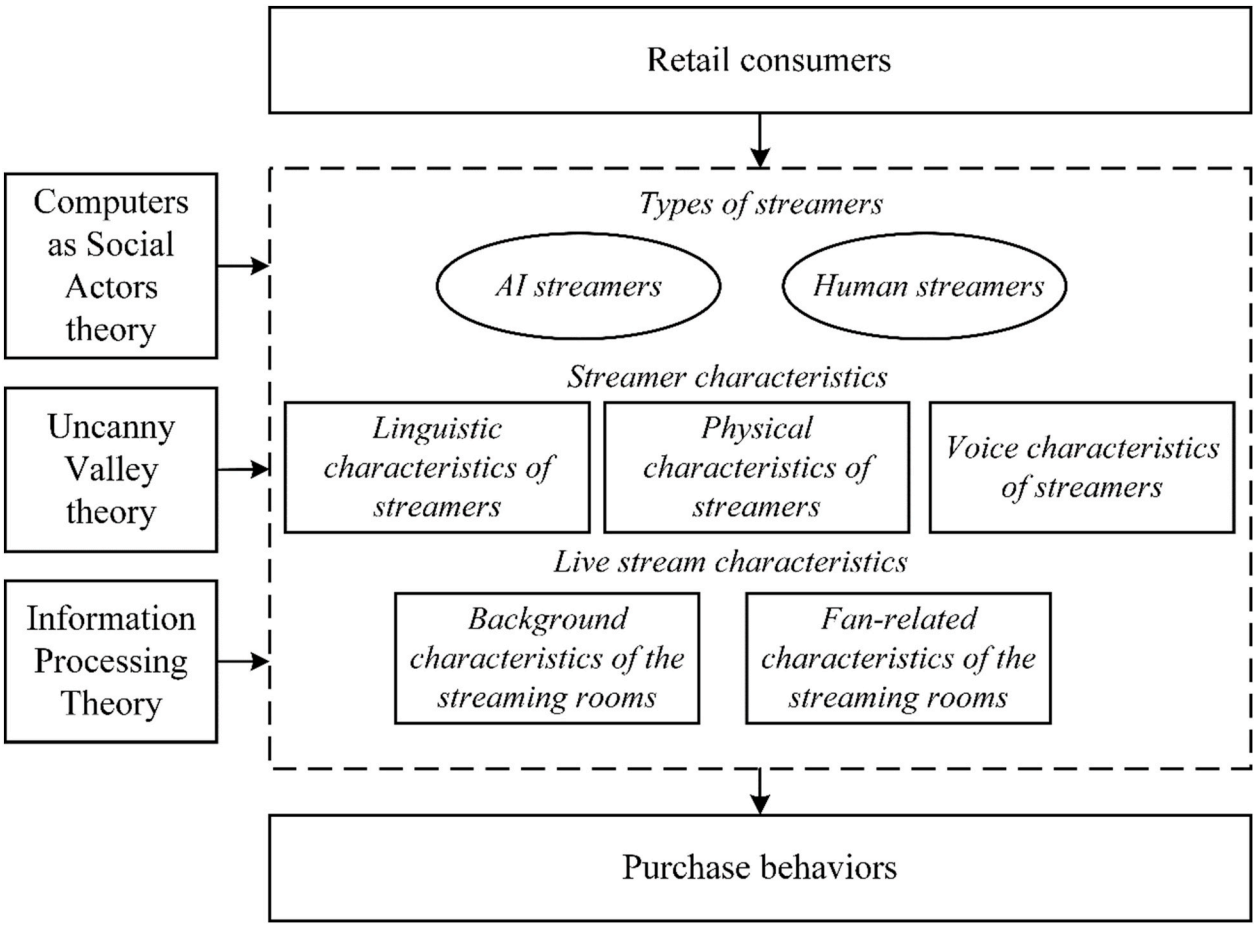


Fig. 1. Theoretical mechanism model.

selected sentiment words and loudness standard deviation as key moderators in our analysis. Sentiment words represent the emotional expressiveness of a streamer's language, which is critical for establishing perceived warmth and engagement in live-streaming contexts (Gong and Sun, 2025). According to the Computers as Social Actors (CASA) framework, emotional language may compensate for the lack of human authenticity in AI streamers, enhancing consumer trust and promoting purchase behaviors. Loudness standard deviation captures the variability in vocal delivery, reflecting human-like vocal dynamics that can influence audience perceptions of authenticity and liveliness. Since AI streamers typically exhibit stable but monotonous speech patterns, introducing controlled variability in loudness can enhance perceived naturalness, thereby improving engagement and purchase intentions (Fu et al., 2024; Wang et al., 2024). By selecting these moderators, we aim to explore how linguistic emotionality and vocal dynamics serve as mechanism pathways that amplify the positive impact of AI streamers on consumer purchasing behaviors, thereby providing a theoretically grounded explanation for the differential effectiveness of AI and human streamers in live e-commerce.

3.2. Linguistic emotionality as a moderator

This section proposes that linguistic emotionality positively moderates the effect of AI streamers on consumer purchase behavior by enhancing perceived social presence. Linguistic emotionality, reflected by the frequency of emotionally expressive words, plays a pivotal role in shaping consumer engagement and purchase behaviors in live-streaming contexts. According to the Computers as Social Actors (CASA) framework, individuals tend to respond to computers and AI agents as social entities when these agents exhibit social cues, such as emotional expressiveness. Emotional language enhances the perceived warmth and social presence of AI streamers, making interactions feel more authentic and engaging (Packard et al., 2024). This perceived human-likeness can reduce psychological distance and increase trust, thereby facilitating consumer purchase decisions during live streams.

In the context of AI streamers, linguistic emotionality may strengthen the effectiveness of AI by compensating for their inherent lack of human warmth and by enhancing the perceived sociability of AI entities (Gong and Sun, 2025). When AI streamers use emotionally expressive language, consumers are more likely to interpret their communication as genuine and engaging, which may amplify the positive effect of AI streamers on consumer purchase behaviors.

H2. *Linguistic emotionality will positively moderate the effect of AI streamers on consumer purchase behaviors.*

3.3. Vocal dynamism as a moderator

This section argues that vocal dynamism positively moderates the effect of AI streamers by enhancing perceptions of human-likeness and reducing robotic monotony. Vocal dynamism, indicated by variability in loudness during speech, contributes to perceptions of human-likeness and social presence in AI-mediated interactions. According to CASA and social presence theories, vocal cues such as loudness variation are critical non-verbal signals that enhance perceived social presence and interactivity during communication (Cascio Rizzo et al., 2023). Vocal monotony is often associated with machine-like interactions, whereas vocal dynamism signals human-like spontaneity and emotional involvement, making AI streamers appear more engaging and less robotic (Huang and Rust, 2022). In the context of live-streaming retail,

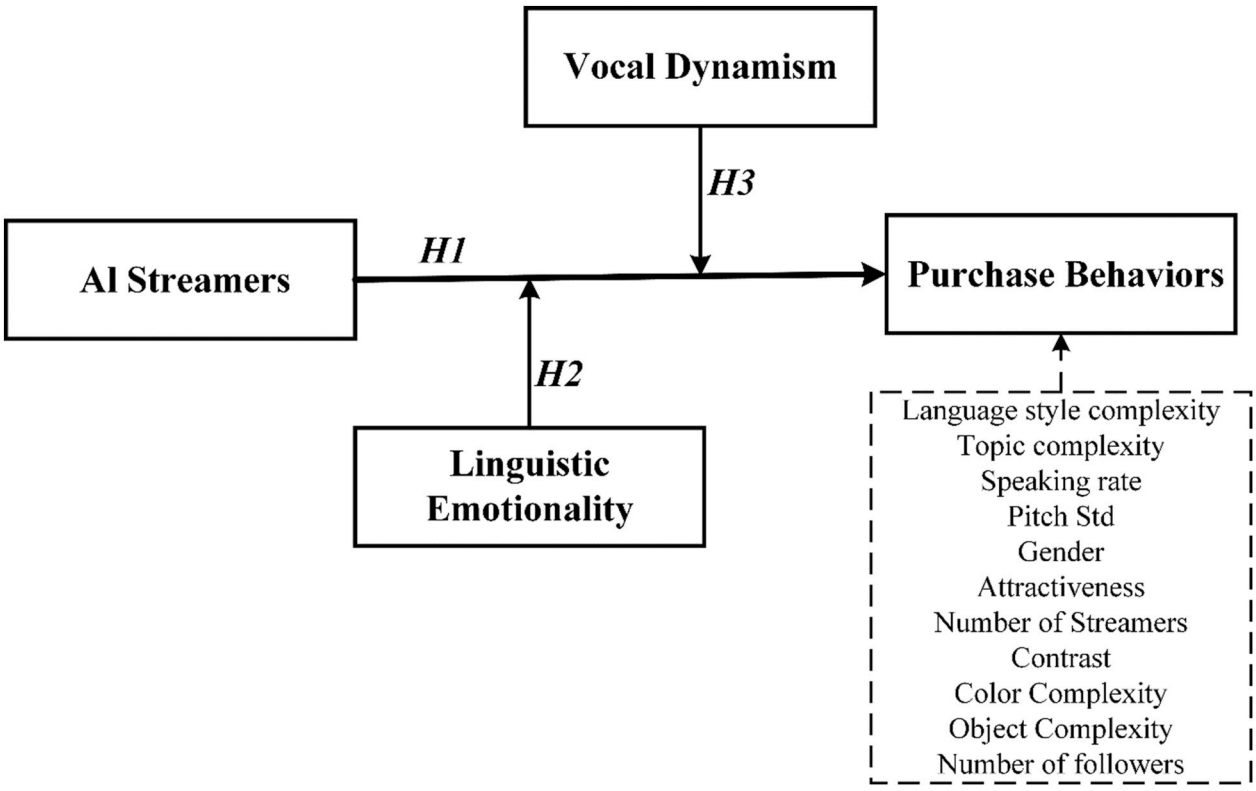


Fig. 2. Framework.

consumers may respond more favorably to AI streamers whose vocal delivery mimics human dynamism, thereby enhancing perceived authenticity and attention during product presentations. This can increase consumer involvement and subsequently lead to higher purchase rates during live streams.

H3. *Vocal dynamism will positively moderate the effect of AI streamers on consumer purchase behaviors*.

4. Data

4.1. Sample selection

Meituan is a prominent Chinese online platform that integrates various local consumer services, including dining, entertainment, and delivery. Currently, it stands as China's most significant local service provider, with approximately 680 million annual transacting users and 9.3 million active merchants.[2] According to data from the China's Takeaway Industry Trend Report, 2025, Meituan held a market share of over 64.6 % in China's food delivery market in 2024, which was more than 2.4 times that of Ele.me (26.4 %), the second-ranked player in the market. Moreover, by implementing subsidy strategies, Meituan achieved a peak daily order volume exceeding 120 million. It operates as an O2O (online-to-offline) service, facilitating user access to local deals through a group-discount model. This platform allows consumers to purchase vouchers from merchants, driving both engagement and commission revenue for Meituan. As one of the world's largest on-demand delivery services, Meituan plays a crucial role in helping users discover merchant information, make informed choices, and complete transactions seamlessly.

This study specifically selected six popular brand stores in Meituan's catering live broadcasts, namely Burger King (fast food restaurant), HaiDiLao (hot pot restaurant), Auntea Jenny (milk tea shop), JuWeiYaBo (braised food store), N-Duo Sushi (Japanese sushi restaurant) and Bliss Cake (cake shop). Each of these six brand restaurants has two live broadcast rooms, one AI live broadcast room and the other real live broadcast room, a total of 12 live broadcast rooms. The data collection time is July 23, 2024. To minimize potential data anomalies due to different live broadcast times, we collected lunch time (3 h, 10:30–13:30), afternoon tea time (2 h, 14:00–16:00), and dinner time (3 h, 16:30–19:30). We preprocessed the original data set to align the live content of the streamer with the user behavior information on the timeline and eliminate redundant, incomplete and invalid data. In addition, non-digital data were converted to digital format to facilitate data analysis. This preprocessing step ensures that the dataset is concise, accurate, and well-suited for subsequent analysis. The final dataset consists of panel data consisting of structured speech texts from 12 live broadcast rooms, each with 480 min of observations, for a total of 5760 min of observations.

Indeed, the selection of six brands on a single platform and day reflects a significant constraint — but also a deliberate methodological choice aimed at preserving the quasi-experimental validity of our identification strategy. In practice, it is rare for a brand to simultaneously operate both an AI livestream room and a human-hosted livestream room under comparable conditions, especially with synchronized time slots and shared product categories. We specifically restricted our sample to such matched pairs in order to minimize endogeneity and sample selection bias. This design allows us to compare AI and human streamers under highly controlled conditions and avoids confounds that would be introduced by comparing across brands or across different days. If we had expanded the sample to include many more retailers who used only AI or only human streamers, the analysis would risk being compromised by non-random adoption — where certain types of firms, products, or consumer bases may systematically favor one type of streamer over another. Such self-selection would

[2] Meituan - statistics & facts. https://www.statista.com/topics/10872/meituan/#topicOverview.

introduce serious endogeneity concerns that are difficult to address convincingly, even with advanced statistical controls.

As shown in Fig. 3, the presentation mode of Meituan Live is the streamer plus a background promotional picture. The AI digital streamer will be marked, and the appearance of the AI digital streamer is completely anthropomorphic and no different from a real person. Fig. 3 shows that the streamer gives marketing feedback to consumers through language interaction. Consumers can click on the shopping cart in the lower right corner to purchase the product, and this data will appear above the comment window.

4.2. Variable operationalizations

4.2.1. Dependent variables and independent variables

Based on the theoretical background and research hypotheses, we selected purchase volume to measure the effectiveness of live-streaming marketing. Analyze the different impacts of AI streamers and human streamers on Purchases. The entire variable description can be found in Appendix Table A1, and the correlation matrix can be found in Appendix Table A2.

The key independent variable is the dummy variable AI, which is 1 for AI streamers and 0 for real streamers. The treatment group in our data is the group with AI streamers, and the control group is the group with human streamers.

4.2.2. Control variables

Linguistic characteristics of streamers: *The Number of Sentiment Words*. We input the text data of each live broadcast room into Kimi (The most popular large language model in Chinese),[3] let Kimi help us analyze the sentiment words in it, and finally give us a sentiment dictionary. We use this sentiment dictionary to calculate the number of sentiment words spoken by the streamers per minute, which represents the streamers' emotional expression. ***Language Style Complexity*.** Referring to Shin et al. (2020), the calculation method is as follows: We use the measurement of information dispersion to calculate the complexity of language style. Specifically, we utilize the Shannon diversity index to measure *Text Language Style Complexity*. As shown in Equation (1), ls_i represents the number of times the *i*-th part of speech words appears in a post divided by the total number of words. ***Topic Complexity*.** As shown in formula (2), we train the LDA (Latent Dirichlet Allocation) model to determine the topic words and calculate the probability associated with each topic tp_i. Subsequently, we apply the Shannon diversity index to quantify *Text Topic Complexity*.

$$LanguageStyleComplexity = -\sum_{i=1}^{n} ls_i \, log(ls_i) \tag{1}$$

$$TopicComplexity = -\sum_{i=1}^{n} tp_i \log(tp_i) \tag{2}$$

Voice characteristics of streamers: *Speaking rate of streamers*. We calculated the number of words spoken by the streamers per unit time. ***The standard deviation of loudness and pitch of streamers' voices*.** We use Python to process the audio and calculate the standard deviation of the loudness and pitch of the streamers' voices per minute.

Physical characteristics of streamers: *Gender characteristics of streamers*. If there is a male streamer, it is 1, otherwise it is 0. ***Attractiveness characteristics of streamers*.** Zhang and Wang (2024) investigated the effect of virtual streamer appearance on purchase intention. We refer to (Luo et al., 2024), the streamer face attractiveness score calculated by Face^{++}.[4]

***Background characteristics of the streaming rooms: Contrast*.** We employ the "Laplace Operator" from Open-Source Computer Vision's image processing method to calculate image *Contrast* (OpenCV: Laplace Operator). ***Color Complexity*.** The calculation of color complexity takes into account the probability of each color appearing in the image and the randomness of these color distributions. We use Shannon Entropy to measure the complexity of color distribution. The *Color Complexity* is determined as depicted in Equation (3), cp_i is the probability of the i-th color appearing. ***Object Complexity*.** We measure the complexity of an image at the overall (*Object Complexity*) following the method by Shin et al. (2020). The *Object Complexity* is determined as depicted in Equation (4), with obj_i denoting the probability of the *i*-th object appearing in the image. To calculate the probability of different objects appearing in an image, we employ BeiT, a cutting-edge neural network in the field of computer vision.

$$ColorComplexity = -\sum_{i=1}^{n} cp_i \, log(cp_i) \tag{3}$$

$$ObjectComplexity = -\sum_{i=1}^{n} obj_i \, log(obj_i) \tag{4}$$

Fan-related characteristics of the streaming room: *The number of followers in the streaming room*. The number of followers is often considered a key factor in assessing a streamer's influence, potential for monetization, and ability to attract new viewers or retain existing ones.

4.3. Descriptive statistics

4.3.1. Descriptive statistics of main variables

Table 2 shows the descriptive statistics of AI live broadcast rooms and real-person live broadcast rooms. From the descriptive statistics of the groups, it can be seen that the mean number of purchases in the AI live broadcast room is higher, and the lower standard deviation means that the purchase volume per minute is more stable. On average, the AI streamers' sentiment word content is higher than that of real people, but their language and topic complexity evaluations are lower than those of real people. For the sound part, the mean of the AI streamers' speech speed, loudness standard deviation, and pitch standard deviation are lower than those of real people, and the AI streamers' voices are softer than those of real people. In terms of appearance attractiveness, the AI streamer is made from digital people, so its appearance attractiveness is higher than that of real people on average. We found that the number of followers in the live broadcast room using AI live broadcast is lower on average than that of real human.

4.3.2. Descriptive statistics of live broadcast rooms

As shown in Table 3, in terms of the number of followers, real human live broadcast rooms usually have more followers than AI live broadcast rooms, except for JuWeiYaBo, whose AI live broadcast room has many more followers than real human live broadcast rooms. In terms of streamer attractiveness, digital streamers in AI live broadcast rooms generally have higher attractiveness scores than real human streamers. The streamers' attractiveness scores of AI live broadcast rooms range from 86 (N-Duo Sushi) to 96.5 (JuWeiYaBo). The streamers' attractiveness scores of real human live broadcast rooms range from 80.75 (HaiDiLao) to 94 (Bliss Cake) (see Table 4).

Fig. 4 shows the comparison of purchases per minute between AI live broadcast rooms and Real Human live broadcast rooms for six catering brands (Burger King, HaiDiLao, Auntea Jenny, JuWeiYaBo, N-Duo Sushi, and Bliss Cake) over 8 h. AI live streaming showed higher purchases, with the two exceptions being Shanghai Auntie and Haidilao, because the gap in the number of followers between their AI live broadcast rooms and Real Human live broadcast rooms was large (Auntea Jenny- AI: 51k, RH: 588k; HaiDiLao- AI: 33k, RH: 326k).

[3] Kimi. https://kimi.moonshot.cn/.

[4] Face++. https://www.faceplusplus.com.cn/beauty/.



Fig. 3. The AI live broadcast room and the real streamer live broadcast room of Burger King on the Meituan platform.

Table 2
Descriptive statistics between AI streamers and human streamers.

	AI Streamers					Real Human Streamers				
VarName	Mean	SD	Min	Median	Max	Mean	SD	Min	Median	Max
Dependent variable										
Purchases	6.419	5.456	0	5	21	6.358	6.85	0	3	22
Linguistic characteristics of streamers										
Sentiment words	46.003	24.157	2	42	125	30.741	16.879	0	28	126
Language style complexity	2.509	0.1	2.174	2.513	2.87	2.539	0.094	1.718	2.543	2.819
Topic complexity	0.857	0.609	0.008	0.809	2.408	0.881	0.33	0.007	0.863	1.817
Voice characteristics of streamers										
Speaking rate	201.582	20.954	38	202	269	222.818	29.93	19	223	329
Loudness Std	795.17	1067.92	32.616	683.405	4168.639	2156.645	1865.73	116.605	1420.452	10080.99
Pitch Std	1005.001	307.025	532.4	910.835	1461.532	1021.316	283.857	313.988	1031.802	1476.135
Physical characteristics of streamers										
Gender	0.167	0.373	0	0	1	0.167	0.373	0	0	1
Attractiveness	91.333	3.739	86	92.25	96.5	88.375	5.091	80.75	89.5	94
Background characteristics of the streaming rooms										
Contrast	0.25	0.029	0.206	0.26	0.282	0.225	0.021	0.196	0.221	0.256
Color Complexity	1.994	0.324	1.456	1.957	2.56	1.738	0.312	1.371	1.722	2.176
Object Complexity	5.88	0.41	5.356	5.748	6.672	6.475	0.601	5.517	6.393	7.36
Fan-related characteristics of the streaming rooms										
Number of followers	11.333	14.842	1.3	5.9	44.2	21.417	20.644	1.3	15.95	58.8

4.3.3. *Descriptive statistics of topics*

The language topic preferences of AI streamers and human streamers should be different (Packard et al., 2024). We implemented the model using Python, with hyperparameters optimized through perplexity minimization. The determination of optimal topic quantity followed a data-driven approach: Using the elbow method on coherence scores, we identified 18 distinct topics as the inflection point where marginal coherence gains diminished (see Fig. 5). These topic variables were subsequently integrated as fixed effects in robustness checks to isolate content-driven heterogeneity.

From Table 5, we find that the AI streamers are counterintuitively designed to be more emotional (e.g., topics 1/4/5), with more keywords

Table 3
Descriptive Statistics Between AI Streaming Rooms and Human Streaming Rooms divided by catering brand stores.

VARIABLES	Type	Number of followers (k)	Gender of Streamers	Attractiveness of Streamers
Burger King	AI	74	female	94
	Real Human	274	female	83.5
HaiDiLao	AI	33	female	91.5
	Real Human	326	male & female	80.75
Auntea Jenny	AI	51	female	87
	Real Human	588	female	86.5
JuWeiYaBo	AI	442	female	96.5
	Real Human	39	female	93
N-Duo Sushi	AI	13	female	86
	Real Human	13	female	92.5
Bliss Cake	AI	67	male	93
	Real Human	45	female	94

Table 4
Topic distribution of AI streams and human streamers.

Topics	Keywords	AI streams	Human streams
1	Babes, Coupons, Set meals, Discounts, Delicious	827	64
2	Link, Live Room, Price, Free, Hours	66	739
3	Live room, Link, Friends, Voucher, Price	98	579
4	Voucher, Delicious, Discount, Taste, Set	606	11
5	Fusion, Taste, Production, Silky, Link	322	134
6	Afternoon, Tea, Links, Live Room, Friends, Taste	227	117
7	Taste, Mouthfeel, Set, High Quality, Babes	47	291
8	Delicious, Dinner, Coupons, Link, Price	178	139
9	Milk tea, Friends, Write off, Link, Pearls	214	70
10	Focus, List, Babes, Followers, Live Room	20	228
11	Friends, Party, Streamer, Hours, Taste	48	192
12	Process, Links, Products, Prices, Vouchers	79	127
13	Coupon, Information, Discount, Refreshing, Food	113	4
14	Yellow, Order, Edible, Link, Shopping bag	35	69
15	Delivery, Package, Delicious, Grapes, Live Room	0	53
16	Sandwich, Product, Original Price, Yellow, Taste	0	49
17	Party, Products, Links, Flavors, Prices	0	8
18	Representative, Set Meal, Friends, Single Item, Type	0	6

such as babes, delicious, and taste. The human streamer' words tend to be more promotional and informative (e.g., topics 2/3), with more about links, vouchers, prices, etc.

4.4. Empirical Models

In the Empirical Models, to test the hypotheses proposed in the theoretical development section, we try to identify the impact of AI streamers on the performance of restaurant live e-commerce. Purchase quantity is a count variable. The non-negative integer nature of our dependent variable calls for a count model, such as the Poisson regression model or a negative binomial regression model. In our dataset, we observed that the variance of Purchase quantity counts exceeds their mean, which is indicative of overdispersion. Therefore, the panel negative binomial model is more appropriate for our data. At the same time, to solve the endogeneity problem caused by omitted variables, we adopt a panel fixed effect model and control the individual fixed effects of the live broadcast room.

Negative binomial panel fixed effects regression eliminates individual effects through conditional likelihood and is suitable for analyzing panel count data with overdispersion. Its core lies in the conditional probability modeling of nonlinear structures, which is essentially different from the within-group deviation method of linear models. These two points enable the negative binomial panel fixed effects model to estimate time-invariant variables. As depicted in Equation (5), the following model is used to estimate the impact of AI Streamer on live e-commerce performance:

$$Purchases_{it} = \beta_0 + \tau_{ATT} AI_{it} + \gamma Controls_{it} + u_i + \nu_t + \varepsilon_{it} \quad (5)$$

In Equation (5), i represents the live broadcast room, and t represents minutes. $Purchases_{it}$ represents the dependent variable, which represents the sales volume of each live broadcast room i in different minutes t. AI_{it} is the core independent variable, which represents whether artificial intelligence technology is used for live broadcast marketing, and β_1 is its estimated coefficient. In addition, *Controls* controls other related factors that affect the sales volume of the live broadcast room, such as the linguistic characteristics of streamers, voice characteristics of streamers, physical characteristics of streamers, background characteristics of the streaming rooms, and other characteristics of the streaming room. u_i represents the control of individual streamer fixed effects, ν_t represents the control of time fixed effects. ε_{it} represents the random disturbance term. The key focus of this study is the coefficient τ_{ATT}. When $\tau_{ATT} > 0$, it means that artificial intelligence technology can improve the performance of live broadcast e-commerce. In the context of this study, the stores set up both real human and AI streaming rooms. The AI streaming rooms were initially operated by AI streamers for marketing purposes, so there is no sample selection bias in this study.

5. Results

5.1. AI streamers and purchases

Table 5 presents the negative binomial regression results examining the impact of AI streamers on purchases, with a particular focus on the moderating roles of sentiment words and loudness standard deviation. Across all models, the coefficient for AI streamers remains significantly positive ($\beta = 2.961$, $p < 0.01$ in Model 1; $\beta = 2.681$, $p < 0.01$ in Model 4), confirming that AI streamers significantly increase the rate of purchase occurrences compared to human streamers, supporting our main hypothesis (H1). This effect is both statistically and practically meaningful—given the dependent variable's count nature, the coefficients correspond to an estimated increase of roughly 14.7- to 19.1-fold in purchase likelihood per minute when switching from human to AI streamers.

We further explore the mechanism pathways by introducing interaction terms: The interaction between AI streamers and sentiment words (AI × Sentiment words) is significantly positive ($\beta = 0.002$, $p < 0.01$ in Models 2 and 4), indicating that the positive impact of AI streamers on purchases is amplified when the streamer's language contains more sentiment-expressive words. This suggests that emotional expressiveness strengthens the effectiveness of AI streamers in driving consumer purchase, providing empirical support for H2. Additionally, the interaction between AI streamers and loudness standard deviation (AI × Loudness Std) is also significantly positive ($\beta = 0.000$, $p < 0.01$ in Models 3 and 4), implying that AI streamers achieve better sales performance when their vocal delivery exhibits greater variability in loudness. This indicates that human-like vocal dynamics enhance the persuasive power of AI streamers during live broadcasts, providing support for H3.

Beyond the main and moderating effects, several control variables demonstrate significant associations with purchase rates, offering further insights into optimizing live-streaming practices: Language style complexity exhibits a negative but non-significant association with

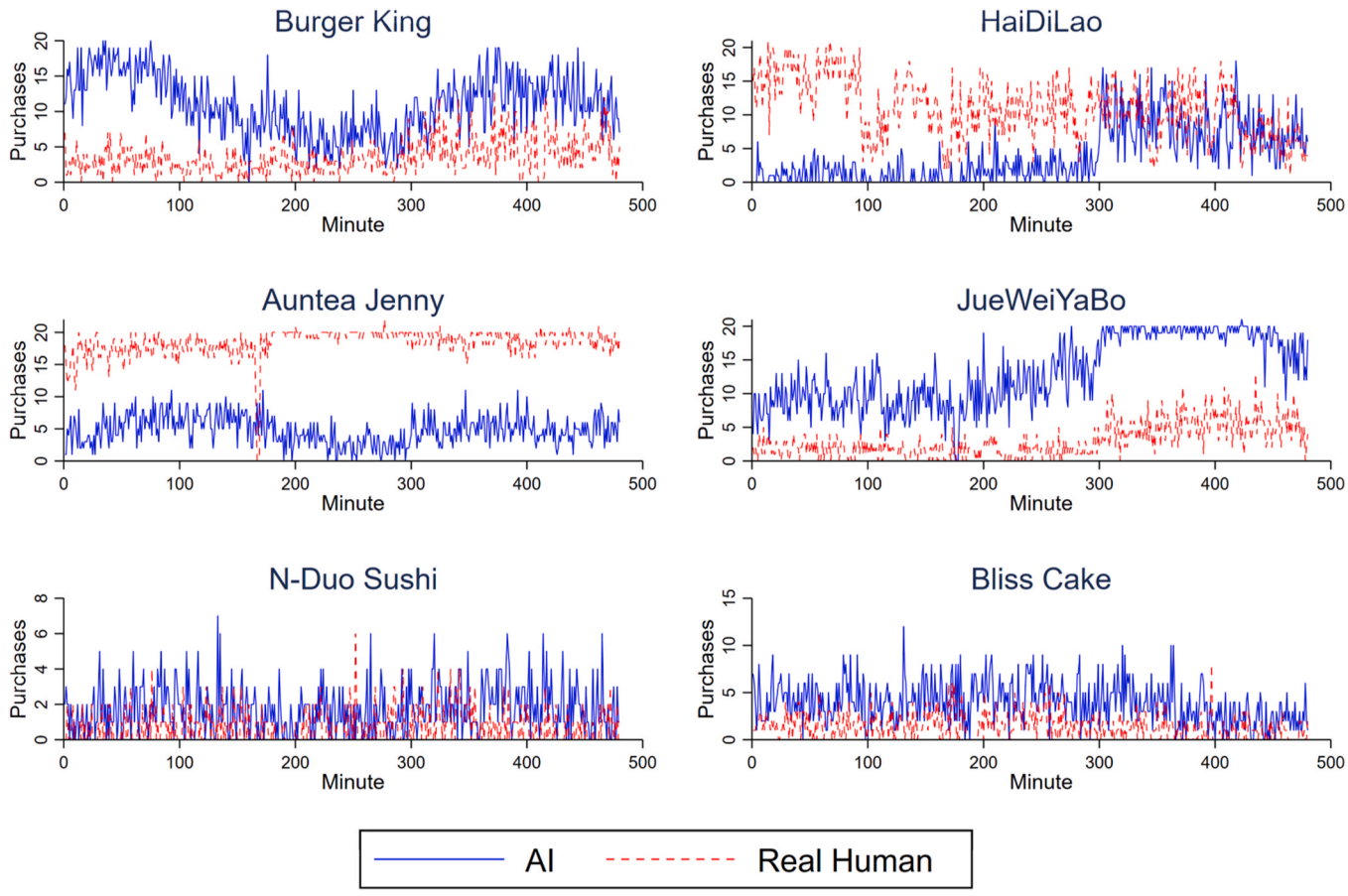


Fig. 4. Performance comparison between AI live broadcast rooms and Real Human live broadcast rooms of six catering brands.

purchases, suggesting a tendency for overly complex language to reduce engagement, although this effect does not reach statistical significance in our models. Topic complexity shows a positive but non-significant relationship with purchase volumes, indicating that discussing moderately complex topics may stimulate interest without overwhelming viewers. Speaking rate has a significant positive association with purchases ($p < 0.01$ across all models), indicating that a faster, yet clear, speaking pace can enhance consumer engagement and prompt impulse purchases during live streams. Loudness standard deviation, when considered independently, has a significant negative association with purchases ($p < 0.01$), suggesting that while overall excessive loudness fluctuations can disrupt viewer experience, controlled variability (as reflected in the positive AI × Loudness Std interaction) can be beneficial. Pitch standard deviation is positively and significantly associated with purchases ($p < 0.01$), implying that human-like variations in pitch enhance the perceived authenticity of streamers and maintain viewer attention.

Gender (male-presenting AI streamer) shows a significant positive impact on purchases ($p < 0.01$), indicating that male-presenting streamers are more effective in the catering live-streaming context. Attractiveness is negatively and significantly associated with purchases ($p < 0.01$), suggesting that excessive focus on streamer appearance may detract attention from the product. Visual background factors consistently show significant negative associations with purchases: Contrast ($p < 0.01$) negatively impacts purchase rates, indicating that overly high contrast may distract viewers during product demonstrations. Color complexity ($p < 0.01$) reduces purchases, emphasizing the value of using clean, non-distracting color schemes. Object complexity ($p < 0.01$) also shows a negative relationship, highlighting the need to minimize unnecessary background clutter during live streams. Collectively, these findings highlight the nuanced interplay between linguistic, vocal, visual, and streamer characteristics in shaping purchase behaviors within live-streaming retail contexts. They provide actionable directions for AI streamer design, live-stream environment optimization, and content strategy refinement.

Overall, these findings illustrate that while AI streamers outperform human streamers in driving purchases, their effectiveness is further enhanced by linguistic emotionality and vocal dynamism, providing clear evidence of the mechanism pathways through which AI streamers influence consumer behavior in live-streaming retail contexts.

5.2. *Robustness checks*

In this study on whether AI can enhance the performance of live-streaming e-commerce, the core dependent variable is the purchases per minute. To ensure the robustness of the findings, two alternative dependent variables are employed: purchases in the following minute and the combined purchases of the current and following minutes. In the context of live-streaming marketing, consumer purchasing behavior often does not occur instantaneously. Most buyers require a brief period to transition from receiving information to making a purchase decision. Thus, a host's marketing rhetoric during the current minute is more likely to have a significant impact on purchases in the following minute, rather than being fully reflected in the current minute's purchases. Furthermore, the influence of marketing rhetoric may extend beyond a single moment, impacting both the current and the subsequent minute. Capturing this cumulative effect provides a more comprehensive understanding of the short-term impact of the host's marketing rhetoric, rather than isolating its effects to a single time frame. By incorporating these two alternative dependent variables, this study examines the impact of marketing rhetoric from multiple perspectives: the first alternative variable (purchases in the following minute) tests the

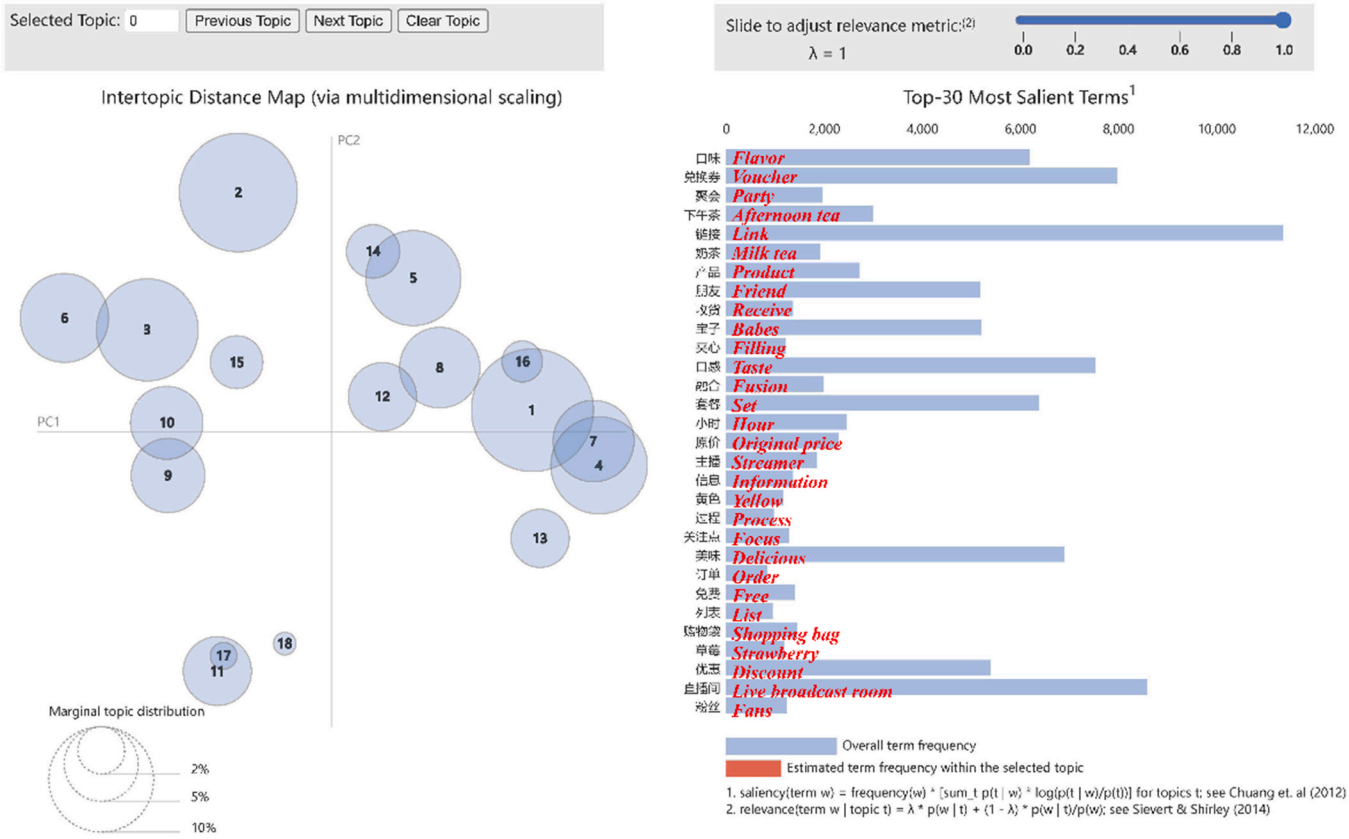


Fig. 5. Performance comparison between AI live broadcast rooms and Real Human live broadcast rooms of six catering brands.

delayed effect of marketing, while the second alternative variable (combined purchases of the current and following minutes) captures its cumulative effect. This robustness test design aligns with the practical dynamics of consumer behavior in live-streaming marketing and mitigates potential biases arising from a single definition of the dependent variable. Consequently, it strengthens the credibility and generalizability of the study's conclusions.

To further eliminate the interference of differences in live content themes on the estimation results, this study incorporates topic fixed effects in the robustness test. The core topics of each minute of live broadcast are identified through the LDA topic model, and the topic fixed effects are controlled to ensure that the model captures the performance differences between AI streamers and real streamers under the same topic. Combining streamer fixed effects (controlling for individual unobservable characteristics) and time fixed effects (excluding time period fluctuations), the triple fixed effects model confirms that the purchase-driving advantage of AI streamers (*Buy*: $B = 2.642$, $p < 0.01$; *Next Buy:* $B = 2.773$, $p < 0.01$; *Total Buy:* $B = 1.988$, $p < 0.01$) comes from their technical capabilities rather than content selection preferences, and the results are highly robust (As shown in Table 6).

6. Conclusions and implications

6.1. Main conclusions

This study explores the impact of AI streamers on the effectiveness of live-streaming marketing within the catering and food delivery sector. Utilizing a quasi-natural experiment from the Meituan platform, we empirically demonstrate that AI streamers significantly enhance live-stream marketing performance. To uncover the mechanistic pathways underlying this impact, we examined the moderating roles of linguistic emotionality and vocal dynamism. Our findings reveal that AI streamers are more effective in driving consumer purchases when their language exhibits higher levels of emotional expressiveness and when their vocal delivery demonstrates greater variability in loudness. These results indicate that emotional language enhances perceived engagement and trust in AI streamers, while human-like vocal variations mitigate perceptions of robotic monotony, thereby increasing consumer involvement and purchase likelihood.

Additionally, the analysis confirms that voice characteristics, including speech speed and pitch variability, play crucial roles in attracting and retaining consumer attention. Faster speech speeds can stimulate impulse purchasing, while vocal dynamism enhances engagement. The study also reaffirms the significance of visual simplicity in live-stream backgrounds, as simpler environments enable consumers to focus on products and interactions without unnecessary distractions. Interestingly, the analysis also reveals a counterintuitive negative relationship between streamer attractiveness and purchase behavior, suggesting that excessive visual perfection may divert consumer attention away from the product and weaken the perceived authenticity of the AI streamer. Collectively, these findings provide valuable theoretical and practical insights into how AI streamers can be leveraged to optimize e-commerce live-stream marketing, emphasizing the importance of linguistic and vocal strategies in enhancing consumer engagement and purchase behaviors during live-streaming sessions.

6.2. Theoretical implications

This study offers novel theoretical contributions to the fields of digital marketing, artificial intelligence, and influencer marketing by empirically validating the positive role of AI streamers in enhancing e-commerce live-streaming effectiveness. By focusing on linguistic emotionality and vocal dynamism as moderating mechanisms, the study advances the Computers as Social Actors (CASA) framework by illustrating how emotional language and human-like vocal variations can enhance perceived social presence and trust in AI-powered influencers.

Table 5
The Impact of AI streamers on Purchases.

Variables	Purchases			
	Model 1	Model 2	Model 3	Model 4
AI × Sentiment words		0.002***		0.002***
		(0.001)		(0.001)
AI × Loudness Std			0.000***	0.000***
			(0.000)	(0.000)
AI	2.961***	2.772***	2.918***	2.681***
	(0.722)	(0.684)	(0.951)	(0.840)
Sentiment words	0.000	−0.001*	0.000	−0.001*
	(0.000)	(0.001)	(0.000)	(0.001)
Language style complexity	−0.071	−0.091	−0.032	−0.052
	(0.085)	(0.085)	(0.084)	(0.084)
Topic complexity	0.017	0.018	−0.001	−0.000
	(0.022)	(0.022)	(0.022)	(0.022)
Speaking rate	0.001***	0.001***	0.001***	0.001***
	(0.000)	(0.000)	(0.000)	(0.000)
Loudness Std	−0.000***	−0.000***	−0.000***	−0.000***
	(0.000)	(0.000)	(0.000)	(0.000)
Pitch Std	0.001***	0.001***	0.001***	0.001***
	(0.000)	(0.000)	(0.000)	(0.000)
Gender	1.510***	1.488***	1.697***	1.672***
	(0.503)	(0.507)	(0.513)	(0.509)
Attractiveness	−0.343***	−0.327***	−0.358***	−0.335***
	(0.087)	(0.084)	(0.116)	(0.106)
Contrast	−41.287***	−39.790***	−38.128***	−35.984***
	(10.332)	(10.214)	(11.502)	(10.916)
Color complexity	−5.680***	−5.566***	−5.886***	−5.700***
	(0.935)	(0.881)	(1.242)	(1.090)
Object complexity	−2.180***	−2.174***	−2.167***	−2.137***
	(0.518)	(0.508)	(0.556)	(0.517)
Number of followers	0.029*	0.032**	0.023	0.027
	(0.015)	(0.015)	(0.018)	(0.018)
Constant	64.880***	62.940***	65.774***	62.650***
	(13.416)	(12.866)	(17.527)	(15.853)
Log-likelihood	−12598.419	−12594.154	−12542.409	−12538.417
Wald chi2	1293.04	1308.67	1418.66	1436.85
Prob > chi2	0.000	0.000	0.000	0.000
Streamer FE	Yes			
Time FE	Yes			
Observations	5760			
Minutes	480			
Streamers	12			

Note: Standard errors in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

Furthermore, drawing on Information Processing Theory (IPT), this study elucidates how the emotional and vocal characteristics of AI streamers influence consumer perceptions, engagement, and purchasing decisions within live-streaming environments. The results underscore the pivotal role of communication dynamics, revealing that language and vocal styles are not merely functional aspects of live-streaming but are central to shaping consumer emotional and cognitive responses in e-commerce contexts. In addition, the analysis identifies a surprising negative association between streamer attractiveness and purchase behavior. This unexpected finding refines existing theoretical understanding by showing that while moderate anthropomorphism can enhance engagement through social heuristics (CASA), excessive attractiveness may cross the affective threshold described by Uncanny Valley theory, leading to reduced credibility and purchase motivation.

In the realm of influencer marketing, this study demonstrates how AI streamers, as a novel form of digital influencers, can effectively engage consumers and drive purchasing behavior through strategically crafted linguistic and vocal interactions. These findings challenge traditional influencer marketing paradigms, suggesting that AI influencers can provide unique advantages over human influencers in terms of consistency, scalability, and personalization, while still fostering perceived human-like engagement through optimized communication strategies. Overall, this study enriches our understanding of e-commerce live-streaming marketing and offers new perspectives for digital marketing practice, particularly in designing AI streamer communication strategies

Table 6
Robustness checks.

VARIABLES	Purchases	Next Purchases	Total Purchases
AI	2.642***	2.773***	1.988***
	(0.611)	(0.640)	(0.248)
Sentiment words	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
Language stylecomplexity	−0.086	−0.058	−0.092
	(0.086)	(0.087)	(0.071)
Topic complexity	0.014	0.007	0.030
	(0.023)	(0.023)	(0.020)
Words	0.001***	0.000	0.001**
	(0.000)	(0.000)	(0.000)
LoudnessStd	−0.000***	−0.000***	−0.000***
	(0.000)	(0.000)	(0.000)
PitchStd	0.001***	0.001***	0.001***
	(0.000)	(0.000)	(0.000)
Gender	1.389***	1.628***	1.350***
	(0.482)	(0.519)	(0.236)
Attractiveness	−0.298***	−0.311***	−0.260***
	(0.079)	(0.078)	(0.026)
Contrast	−37.04***	−40.06***	−32.90***
	(9.700)	(10.27)	(3.316)
Color_complexity	−5.246***	−5.419***	−4.275***
	(0.781)	(0.832)	(0.274)
Object_complexity	−1.993***	−2.135***	−1.603***
	(0.461)	(0.515)	(0.205)
Number of followers	0.037**	0.032**	0.019***
	(0.017)	(0.015)	(0.005)
Constant	57.51***	61.06***	49.41***
	(11.92)	(12.33)	(3.651)
Streamer FE	Yes		
Time FE	Yes		
Topic FE	Yes		
Log-likelihood	−12591.159	−12581.122	−15599.011
Wald chi2	1329.64	1287.83	2024.71
Prob > chi2	0.000	0.000	0.000
Observations	5760	5748	5748
Minutes	480	479	479
Streamers	12		

Note: Standard errors in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

that enhance consumer engagement and reshape influencer marketing in the era of artificial intelligence specifically within the context of high-frequency O2O services like food delivery.

6.3. Practical implications

This study offers practical insights for practitioners in the e-commerce and digital retail sectors seeking to integrate AI technology into live-stream marketing. The findings demonstrate that AI streamers can significantly enhance consumer engagement and purchase behaviors when their communication strategies are carefully designed to leverage emotional expressiveness and human-like vocal dynamism. First, businesses should prioritize the design of linguistic strategies for AI streamers by incorporating emotionally expressive yet concise language to foster consumer connection and engagement. Avoiding overly complex language while maintaining an optimal level of topic complexity can help sustain viewer attention and drive impulse purchases during live streams.

Second, companies should focus on vocal strategies, ensuring that AI streamers maintain a moderate speaking rate while incorporating human-like variations in pitch and tone. Minimizing excessive fluctuations in loudness while maintaining vocal dynamism can help AI streamers simulate human-like interactivity, thereby enhancing the perceived authenticity of the live-streaming experience. Third, the study underscores the importance of visual and background simplicity during live streaming sessions. Using clean, high-contrast backgrounds can reduce distractions, allowing viewers to focus on the product and the streamer's interactions, thereby fostering a more engaging and seamless shopping experience. Fourth, regarding appearance and gender design,

businesses should tailor the gender and aesthetic design of AI streamers to align with their target demographics while ensuring consistency with brand identity. Maintaining a balanced level of attractiveness can help avoid unnecessary distractions, particularly in food and beverage live-streaming contexts, where overemphasis on appearance may detract from product focus.

Table 7 summarizes these strategies for optimizing AI streamer performance based on our empirical findings, covering linguistic, vocal, visual, and design factors. By leveraging these actionable insights, businesses can refine their live-streaming marketing strategies, enhancing customer engagement, purchase behaviors, and overall live-streaming performance in the digital retail landscape.

From a managerial perspective, this study offers actionable guidance for platform operators and brand managers seeking to leverage AI streamers effectively and responsibly. First, the findings suggest that AI streamers can outperform human hosts when their linguistic, vocal, and visual cues are carefully calibrated — emotional expressiveness should humanize rather than exaggerate, while moderate vocal and visual variation maintains authenticity and viewer comfort. Second, to ensure AI transparency, platforms should clearly disclose the non-human identity of AI streamers and provide users with opt-in or labeling mechanisms, thereby avoiding potential consumer deception and maintaining trust. Ethical design also entails algorithmic fairness — for instance, preventing bias in attractiveness scoring or voice tone calibration that might favor certain gender or appearance types. Furthermore, platforms could implement AI performance dashboards that allow brand managers to fine-tune linguistic sentiment, vocal tone, and visual settings based on real-time analytics. Retailers might also conduct A/B tests comparing AI and human streamers across product types and audience segments to identify conditions under which AI presence is most effective. Finally, firms adopting AI streamers should complement automation with human oversight, ensuring that customer feedback, cultural sensitivity, and ethical standards remain central to AI deployment. Collectively, these strategies support both the commercial scalability and responsible governance of AI applications in live commerce.

6.4. Limitations and future direction

This study focused on the catering industry within China's Meituan platform, which limits the generalizability of the findings. Differences in other industries and market dynamics may lead to varying impacts on the effectiveness of AI streamers in live streaming. Additionally, due to data acquisition limitations, this study only examined 6 brands and 12 live broadcast rooms, suggesting that future research should expand the scope to include a larger variety of industries, brands, and sample sizes. It is also important to note that revenue data, which could provide further insights into the economic impact of AI streamers, was unavailable for this study. This limitation should be acknowledged, as the lack of such data restricts a comprehensive understanding of the direct financial outcomes associated with AI streamers in live streaming contexts. As AI streamers continue to gain popularity, consumer acceptance and trust in AI technology may evolve, potentially affecting the long-term validity of the research conclusions. Future research could explore these dynamics over time. Recent cross-cultural studies in online retailing and branding (Hyun et al., 2025; Park et al., 2025; Park and Ahn, 2025) show that consumer motives, brand values, and emotional attachment vary significantly across markets, highlighting the importance of extending AI streamer research to diverse cultural and industry contexts.

While the current study draws on behavioral data from Meituan, China's leading O2O platform, the underlying mechanisms identified—namely the roles of linguistic emotionality, vocal dynamism, and visual simplicity in driving consumer engagement—are theoretically generalizable across digital retail environments. Nonetheless, we acknowledge that platform algorithms, consumer norms, and cultural expectations may vary across markets. Future research could extend

Table 7
Guidelines for optimizing AI streamer performance based on empirical results.

Factor	Variables	Optimal Strategy	Details/Explanation
Linguistic Strategies	Sentiment words	Use emotionally expressive language	Emotionally expressive language is positively associated with purchases. AI streamers should incorporate enthusiastic, emotionally resonant words to enhance engagement and drive impulse purchases.
	Language style complexity	Use simple, clear language	Higher language complexity reduces purchases. Avoid overly complex or formal language, using clear and accessible expressions to maintain audience attention and comprehension.
	Topic complexity	Engage with moderately complex topics	Topic complexity positively correlates with purchases. Discussing moderately complex topics can stimulate interest and engagement, encouraging consumers to stay and purchase.
Vocal Strategies	Speaking rate	Maintain a faster, but clear, speaking rate	Faster speaking rates are associated with higher purchases. AI streamers should adopt an energetic yet clear speaking pace to encourage impulse buying without overwhelming viewers.
	Loudness Std	Incorporate vocal dynamism	Greater variations in loudness positively correlate with purchases. Introducing human-like loudness variations enhances authenticity and viewer engagement while avoiding monotony.
	Pitch Std	Vary pitch and tone	Pitch variability positively influences purchases. AI streamers should use natural pitch and tone fluctuations to maintain interest and simulate human-like interaction.
Appearance and Gender Design	Gender	Prefer male-presenting AI streamers	Male-presenting streamers are positively associated with higher purchases in catering live streams. Gender presentation should align with audience preferences while matching brand positioning.
	Attractiveness	Avoid excessive focus on attractiveness	Attractiveness negatively correlates with purchases in catering live streams. Maintain a balanced, brand-aligned appearance without overemphasizing physical attractiveness to keep focus on products.
Visual & Background Strategies	Contrast	Use moderate, non-distracting contrast	Higher background contrast reduces purchases. Avoid overly

(continued on next page)

Table 7 (*continued*)

Factor	Variables	Optimal Strategy	Details/Explanation
			stark contrasts and instead use balanced visuals to keep focus on products while maintaining viewer comfort.
	Color complexity	Use simple, non-complex color schemes	Higher color complexity reduces purchases. Utilize clean, consistent color schemes in backgrounds to minimize distractions and improve the shopping experience.
	Object complexity	Maintain minimal background clutter	Background object complexity negatively impacts purchases. Use simple, clear backgrounds without unnecessary objects to help consumers maintain focus on products and interactions.

these findings by replicating the AI–Human Complementarity Framework in other contexts such as TikTok Live, Taobao Live, or Western live commerce platforms, as well as across different product categories (e.g., apparel, beauty, electronics). Such cross-market validations will help establish the robustness and boundary conditions of AI streamer effectiveness beyond the Meituan setting.

CRediT authorship contribution statement

Yuzhou Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yulin Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Haowen Fan:** Writing – review & editing, Conceptualization. **Yiting Guo:** Writing – review & editing, Validation, Software, Methodology, Formal analysis.

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Declaration of competing interest

The authors have no conflict of interest.

Appendix A. *Variable operationalizations*

Table A1
Variable operationalizations

Variable	Method	Detail
Purchases	Counting	We calculated the purchase volume in the live broadcast room per unit time.
AI	Observe AI Live Room Identifier	AI streamers live broadcast will have an Identifier.
Sentiment words	Large Language Model (Kimi)	We input the text data of each live broadcast room into Kimi (The most popular large language model in Chinese), let Kimi help us analyze the sentiment words in it, and finally give us a sentiment dictionary. We use this sentiment dictionary to calculate the number of sentiment words spoken by the streamers per minute, which represents the streamers' emotional expression.
Language Style Complexity	Shannon Diversity Index	We use the measurement of information dispersion to calculate the complexity of language style. Specifically, we utilize the Shannon diversity index to measure Text Language Style Complexity
Topic complexity	LDA model	We train the LDA (Latent Dirichlet Allocation) model to determine the topic words and calculate the probability associated with each topic. Subsequently, we apply the Shannon diversity index to quantify Text Topic Complexity.
Speaking rate	Counting	We calculated the number of words spoken by the streamers per unit time.
Loudness Std Pitch Std	Audio Analysis	We use Python to process the audio and calculate the standard deviation of the loudness and pitch of the streamers' voice per minute.
Contrast	OpenCV: Laplace Operator	We employ the "Laplace Operator" from Open-Source Computer Vision's image processing method to calculate image *Contrast* (OpenCV: Laplace Operator).
Object complexity	Neural Network	To calculate the probability of different objects appearing in an image, we employ BeiT, a cutting-edge neural network in the field of computer vision.
Color complexity	Deep Learning	The calculation of color complexity takes into account the probability of each color appearing in the image and the randomness of these color distributions.
Gender	Observe	
Attractiveness	Deep Learning	Self-developed deep learning framework "MegEngine" from MEGVII. (Face++. https://www.faceplusplus.com.cn/beauty/)
Number of followers	Observe	Observe the number of followers of 12 live broadcast rooms at the same time

Appendix B. *Correlation Matrix*

Table A2
Correlation Matrix

	Purchases	AI	Sentiment words	Language Style complexity	Topic complexity	Speaking rate	Loudness Std	Pitch Std	contrast	Object complexity	Color complexity	gender	Attractiveness	Number Of followers
Purchases	1	0.09***	0.04***	−0.02	0.13***	0.02	−0.29***	0.09***	0.06***	0.12***	−0.21***	0.15***	−0.01	0.74***
AI	0.005	1	0.32***	−0.15***	−0.07***	−0.41***	−0.56***	−0.02*	0.39***	−0.48***	0.24***	0.00	0.29***	−0.12***
Sentiment words	0.04***	0.34***	1	0.15***	−0.25***	0.04***	−0.02	−0.34***	0.13***	−0.25***	0.01	−0.10***	0.33***	−0.08***
Language Style Complexity	0.02*	−0.15***	0.14***	1	−0.22***	0.04***	−0.08***	−0.36***	0.01	−0.31***	−0.12***	−0.32***	0.23***	−0.06***
Topic complexity	0.06***	−0.02*	−0.24***	−0.22***	1	0.06***	0.09***	0.57***	0.08***	0.48***	−0.29***	0.42***	−0.15***	0.33***
Speaking rate	0.05***	−0.38***	0.01	0.03*	0.10***	1	0.43***	0.04***	0.04***	0.44***	0.07***	0.42***	−0.26***	0.15***
Loudness Std	−0.24***	−0.41***	−0.12***	−0.12***	0.20***	0.35***	1	0.13***	−0.05***	0.49***	−0.02	0.40***	−0.10***	−0.06***
Pitch Std	0.03**	−0.03**	−0.35***	−0.34***	0.56***	0.00	0.23***	1	0.27***	0.61***	−0.26***	0.49***	−0.17***	0.31***
contrast	−0.13***	0.45***	0.09***	−0.00	0.17***	−0.03**	−0.00	0.25***	1	−0.09***	0.03***	0.19***	0.23***	0.15***
Object complexity	0.16***	−0.50***	−0.31***	−0.28***	0.44***	0.40***	0.47***	0.60***	−0.14***	1	−0.34***	0.58***	−0.23***	0.42***
Color complexity	−0.25***	0.37***	0.08***	−0.17***	−0.24***	0.05***	0.00	−0.19***	0.06***	−0.29***	1	0.19***	−0.58***	−0.42***
gender	0.08***	0.00	−0.12***	−0.33***	0.51***	0.41***	0.32***	0.48***	0.16***	0.63***	0.24***	1	−0.23***	0.26***
Attractiveness	−0.10***	0.31***	0.34***	0.24***	−0.14***	−0.26***	−0.24***	−0.29***	0.25***	−0.33***	−0.38***	−0.28***	1	−0.04***
Number of followers	0.75***	−0.27***	−0.16***	−0.00	0.20***	0.23***	−0.01	0.12***	−0.13***	0.39***	−0.38***	0.08***	−0.29***	1

Note: Lower-triangular cells report Pearson's correlation coefficients. Upper-triangular cells are Spearman's rank correlation. ***p < 0.01, **p < 0.05, *p < 0.1.

Appendix C. *Pearson Correlations Between Sentiment Scores and Validation of AI-Based Sentiment Word Method*

This appendix reports the results of a Pearson correlation analysis that compares the sentiment indicators generated by the AI-based sentiment word method (Kimi) used in this study with three widely adopted sentiment analysis tools—SnowNLP, HowNet,[5] and Pysenti[6] (Zhang et al., 2025). The purpose is to verify the accuracy and appropriateness of the Kimi-based approach for the catering live-streaming context by evaluating the correlation strength and statistical significance across methods.

For each text segment, we calculated sentiment-related indicators using the four methods while ensuring a consistent analysis unit (per minute) and identical data formatting: **Kimi** (AI-based sentiment word method, this study): The number of sentiment words per minute (sum of positive and negative sentiment words generated by Kimi's catering-specific dictionary, with positive words assigned +1 and negative words assigned −1 to form a "net sentiment word count" for correlation calculation). **SnowNLP**: Sentiment score per minute (range: 0–1, with higher values indicating stronger positive sentiment; calculated by averaging the sentiment scores of all sentences in the segment). **HowNet**: Semantic sentiment score per minute (based on OpenHowNet's semantic annotation, words with "positive evaluation" sememes were assigned +1, "negative evaluation" sememes −1, and neutral words 0, then averaged by word count). **Pysenti**: Rule-based sentiment word count per minute (using Pysenti's rule mode, net sentiment word count was calculated by +1 for positive words and −1 for negative words).

Pearson correlation coefficients were calculated to assess the linear relationships between Kimi's indicator and the three benchmarks. Table A3 presents the coefficients, standard errors, and significance levels. All coefficients are positive and statistically significant at the 0.01 level, confirming that the Kimi method aligns with the sentiment-assessment logic of mainstream tools and correctly captures high-versus-low sentiment differences.

Table A3
Pearson Correlations Between Sentiment Scores

	Kim	SnowNLP	HowNet	Pysenti
Kim	1			
SnowNLP	0.135***	1		
HowNet	0.403***	1	1	
Pysenti	0.265***	0.406***	0.260***	1

Note: Lower-triangular cells report Pearson's correlation coefficients.

All correlations between Kimi and the three comparison tools are positive (r = 0.135–0.403). This indicates that when Kimi detects a higher net sentiment word count (i.e., more positive expression) in a live-stream segment, SnowNLP (continuous sentiment score), HowNet (semantic score), and Pysenti (rule-based count) also classify the same segment as more positive. No directional inconsistencies (e.g., Kimi positive vs. others negative) are observed, confirming the conceptual alignment of Kimi's catering-specific sentiment recognition with mainstream methods. In summary, the AI-based sentiment word approach (Kimi) demonstrates strong reliability and conceptual validity for the catering live-streaming scenario and provides a sound empirical basis for subsequent quantitative analyses in this study.

[5] https://openhownet.thunlp.org/.
[6] https://pypi.org/project/pysenti/.

Appendix D. Pearson Correlations Between AI-Based Attractiveness Scores and Validation of Face++ Method

To mitigate potential subjectivity bias from relying on a single AI tool and to verify whether Face++ accurately captures general perceptions of facial attractiveness, we conduct a cross-validation using two additional AI-based attractiveness scoring platforms. We selected two additional mainstream AI attractiveness scoring platforms for cross-validation: **Face++**[7]: The primary tool in the main regression, developed by Megvii. It uses advanced computer vision algorithms to extract facial features (e.g., symmetry, proportion, skin texture) and outputs a standardized attractiveness score (0–100, higher scores indicate stronger perceived attractiveness). It is widely applied in academic research and industrial scenarios for facial attribute analysis (Luo et al., 2024). **Beauty Score Test**[8]: Labeled "AI 2″ in this appendix. It is an AI-based attractiveness scoring tool that processes images on the server side and outputs a score on a 1–10 scale (Alkusari et al., 2024). Although its specific algorithm is not publicly disclosed, it is validated in prior studies for consistency with general attractiveness perceptions. **Attractiveness Test**[9]: Labeled "AI 1″ in this appendix. It uses **deep neural networks** trained on hundreds of thousands of human-annotated facial images (supervised learning framework) to predict attractiveness. It focuses on universal facial beauty features (e.g., facial symmetry, golden ratio) and outputs a 1–10 score (Alkusari et al., 2024).

Table A4 presents Pearson correlation coefficients among the three AI tools. All correlations are positive and statistically significant, confirming that Face++ scores are moderately to strongly correlated with both Attractiveness Test and Beauty Score Test, indicating a consistent ranking pattern across different algorithms. The strong positive correlations (r = 0.64–0.70) indicate that all tools produce comparable attractiveness rankings, implying that Face++ reflects widely shared aesthetic judgments rather than idiosyncratic algorithmic bias.

Table A4
Pearson Correlations Between AI-Based Attractiveness Scores

	Face++	Beauty score test	Attractiveness test
Face++	1		
Beauty score test	0.640**	1	
Attractiveness test	0.696**	0.915***	1

Note: Lower-triangular cells report Pearson's correlation coefficients.

Correlation alignment confirms consistency in relative ranking, but empirical validity requires demonstrating that Face++ scores meaningfully explain live-streaming purchase behavior. We therefore replicate the main negative-binomial regression using attractiveness scores from each tool as the independent variable while keeping all model specifications identical. Table A5 summarizes the results. Across models, the coefficients of AI and other covariates remain consistent, and the Face++ model shows similar or slightly stronger explanatory power compared with the alternatives, confirming its robustness and external validity.

Table A5
The Impact of AI streamers on Purchases

Variables	Model 1	Model 2	Model 3
	Face++	Beauty score test	Attractiveness test
AI	2.961***	2.007***	2.091***
	(0.722)	(0.5878)	(0.5760)
Attractiveness	**−0.343*** **	**−0.023***	**−0.013**
	(0.087)	**(0.0123)**	**(0.0085)**
Sentiment words	0.000	0.000	0.000
	(0.000)	(0.0004)	(0.0004)
Language style complexity	−0.071	−0.073	−0.073
	(0.085)	(0.0851)	(0.0851)
Topic complexity	0.017	0.012	0.011
	(0.022)	(0.0218)	(0.0218)
Speaking rate	0.001***	0.001***	0.001***
	(0.000)	(0.0003)	(0.0003)
Loudness Std	−0.000***	−0.000***	−0.000***
	(0.000)	(0.0000)	(0.0000)
Pitch Std	0.001***	0.001***	0.001***
	(0.000)	(0.0001)	(0.0001)
Gender	1.510***	0.486	0.670
	(0.503)	(0.4327)	(0.4235)
Contrast	−41.287***	−23.995***	−22.662***
	(10.332)	(7.5824)	(7.4612)
Color complexity	−5.680***	−4.053***	−4.154***
	(0.935)	(0.7073)	(0.6959)
Object complexity	−2.180***	−1.871***	−1.970***
	(0.518)	(0.4982)	(0.4869)
Number of followers	0.029*	0.146***	0.151***
	(0.015)	(0.0337)	(0.0345)
Constant	64.880***	25.759***	25.665***
	(13.416)	(5.0199)	(4.9661)
Log-likelihood	−12598.419	−12606.101	−12606.636
Wald chi2	1293.04	1352.49	1351.48

(*continued on next page*)

[7] https://www.faceplusplus.com.cn/.
[8] https://www.beautyscoretest.com/zh/.
[9] https://attractivenesstest.com/.

Table A5 (*continued*)

Variables	Model 1	Model 2	Model 3
	Face++	Beauty score test	Attractiveness test
Prob > chi2	0.000	0.000	0.000
Streamer FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	5760	5760	5760
Minutes	480	480	480
Streamers	12	12	12

Note: Standard errors in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

Taken together, these results verify that Face++ provides credible and empirically valid attractiveness measurements. Its performance is robust across alternative AI methods and can be confidently interpreted as reflecting audience-perceived streamer attractiveness in live-commerce environments. An unexpected yet theoretically meaningful finding of this study is the negative relationship between AI streamer attractiveness and purchase volume. While prior research often links attractiveness to persuasion effectiveness, our results suggest that in the AI context, higher visual perfection may trigger perceptual distance rather than affinity. Excessively idealized AI avatars can induce a "perceived artificiality" effect, reducing the sense of authenticity and credibility. From an information-processing perspective, overemphasized visual cues may also displace cognitive attention away from product information, weakening utilitarian evaluation and purchase motivation. This phenomenon aligns with the Uncanny Valley framework, indicating that the persuasive power of AI agents follows an inverted-U pattern with respect to anthropomorphism and visual realism. A moderate level of attractiveness, where AI agents remain appealing yet recognizably artificial, appears most conducive to maintaining both viewer comfort and engagement. This interpretation not only contextualizes the negative coefficient observed in the empirical analysis but also offers actionable insights for platform designers in optimizing AI avatar presentation.

Data availability

Data will be made available on request.

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